(PRACTICE) Midterm, Math 170S, Winter 2020
Instructor: Elizaveta Rebrova

Printed name: ____

Signed name: ____

Student ID number: _____________________________________

Instructions:

- Read problems very carefully. If you have any questions please ask.
- **Do not forget to show all your work.** The correct final answer alone is not sufficient for full credit - you should explain your answers as much as you can, saving enough time to attempt all the problems.
- You final answers does not need to be completely simplified (unless otherwise stated), e.g. a sum of several decimal fractions would be completely fine. Ask if in doubt.
- Books, calculators, phones are not allowed, please put them away. If you need more paper please raise your hand.

Question	Points	Score
1	15	
2	15	
3	10	
4	10	
5	10	
Total:	60	

1. (a) (3 points) Let $X_1, X_2, \ldots, X_{20}$ are samples taken from normal $N(\mu_X, 2)$ distribution $(2 = \sigma^2)$, $Y_1, Y_2, \ldots, Y_{40}$ are samples taken from normal $N(\mu_Y, 4)$ distribution, $Z_1, Z_2, \ldots, Z_{45}$ are taken from normal $N(\mu_Z, 5)$ distribution.

   All $X_i's$, $Y_i's$ and $Z_i's$ are mutually independent. Additionally, you computed that What is the distribution of a random variable $5\overline{X} - \overline{Y} - 3\overline{Z}$?

   For full credit you should either give a name of the distribution and compute its parameters; or compute a density function.

   Now we additionally computed that

   $X_1 + \ldots + X_{20} = 1;$

   $Y_1 + \ldots + Y_{40} = 10;$

   $Z_1 + \ldots + Z_{45} = 3.$

   (b) (2 points) Give a point estimate for a parameter $5\mu_X - \mu_Y - 3\mu_Z$.

You must give numerical answers; but you do not need to simplify them completely, e.g., a sum of several fractions would be completely fine

(c) (5 points) Find 2-sided 95% confidence interval for a parameter $5\mu_X - \mu_Y - 3\mu_Z$.

(d) (5 points) Find 1-sided 80% confidence interval for the upper bound of the parameter $5\mu_X - \mu_Y - 3\mu_Z$.

2. We observe a data sample $X_1, \ldots, X_n$.

   Let $f(x; \theta) = (1/\theta)x^{(1/\theta)-1}$, $0 < x < 1$, $0 < \theta < \infty$.

   (a) (5 points) Find $\hat{\theta}$ = the maximum likelihood estimate of θ.

   (b) (5 points) Find $\tilde{\theta}$ = the method of moments estimate of θ.

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(c) (5 points) (a) Is $\hat{\theta}$ an unbiased estimate of θ? (b) Is $\tilde{\theta}$ an unbiased estimate of θ?

For (b), you may use without the proof that $\mathbb{E}(1/\overline{X}) \neq 1/\mathbb{E}(\overline{X})$

3. (10 points) Suppose that observations are coming from a distribution with the density function

$$f_X(x) = \theta e^{-\theta x} \quad \text{for} \quad x \geq 0,$$

where θ is an unknown parameter. Additionally you know that θ is either $1, 2$ or 3. Prior the experiment, you have no idea which option is more likely (so, let's assume that all options are equally likely). Then you observe in 3 experiments that $X = 3, 2, 5$. Find the MAP estimator of Θ. Give numerical value as an answer.

4. (10 points) We have two samples X_1, X_2 taken from the uniform distribution

$$Unif([\theta - 2, \theta + 1]).$$

What is the length of a two-sided confidence interval with confidence level 80% for the estimate $X_{(2)}$ for parameter θ? $X_{(2)}$ denotes the second order statistic.

5. (10 points) Let $X_1, X_2, \ldots, X_n$ be samples taken from some distribution with unknown density function. However, you know that $\mathbb{E}X = \alpha/\beta$ and $\operatorname{Var} X = \alpha/\beta^2$, where α and β are two parameters of the density function (and there is no third parameter). What method could we use to find good estimates for α and β? What are these estimates?